NPC – Unit 3 Assignment

Topic: Configure a DNS Server and Web Server Using Cisco Packet Tracer

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Course: Network Protocol Configuration (Fortinet)

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1. Aim

To design and configure a computer network using Cisco Packet Tracer, configure a Domain Name System (DNS) server and a web server, assign IP addresses to the network devices, and verify connectivity and domain name resolution.

2. Objectives

1. To create a network topology using PCs, a switch, a DNS server, and a web server.
2. To assign appropriate IP addresses to all devices.
3. To configure the DNS service and create a domain name record.
4. To enable the HTTP service on the web server.
5. To configure client PCs to use the DNS server.
6. To verify network connectivity using the ping command.
7. To test domain name resolution and access the website using its domain name.

3. Software and Requirements

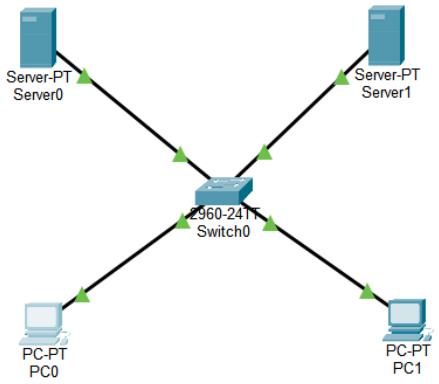
Sr. No.	Component	Purpose
1	Cisco Packet Tracer	Network simulation and configuration
2	Two PCs	Client devices used for testing
3	One Cisco 2960 switch	Connects all network devices
4	One DNS server	Resolves domain names into IP addresses
5	One web server	Hosts the website
6	Copper straight-through cables	Connects the devices to the switch

4. Network Topology

The network consists of two client PCs, one switch, one DNS server, and one web server. All devices are connected to the switch and configured within the same IP subnet, 192.168.1.0/24.

The DNS server is responsible for resolving the domain name www.mimi.com to the IP address of the web server. The web server hosts the webpage, which can be accessed by the client PCs through a web browser.

Screenshot 1: Complete network topology in Cisco Packet Tracer



5. IP Addressing Scheme

The following IP addressing scheme is used for the network.

Device	Interface	IP Address	Subnet Mask	DNS Server
Server0 (DNS)	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.2
Server1 (Web)	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.2
PC0	FastEthernet0	192.168.1.10	255.255.255.0	192.168.1.2
PC1	FastEthernet0	192.168.1.11	255.255.255.0	192.168.1.2

All devices belong to the same subnet, so a default gateway is not required for communication within this local network.

The DNS server address is configured as 192.168.1.2 on the client PCs and the web server so that they can use the DNS service.

6. Configuration Procedure

6.1 Creating the Network Topology

1. Open Cisco Packet Tracer.
2. Place two PCs, one Cisco 2960 switch, and two servers in the workspace.
3. Rename the first server as the DNS server and the second server as the web server, if required.
4. Connect all four end devices to the switch using copper straight-through cables.
5. Wait until the connection indicators turn green.

6.2 Configuring the IP Addresses

1. Open each PC or server.
2. Navigate to **Desktop → IP Configuration**.
3. Select static IP configuration.
4. Enter the IP address and subnet mask according to the IP addressing table.
5. Enter 192.168.1.2 as the DNS server address.

The devices are now configured with unique IP addresses in the same subnet.

6.3 Configuring the DNS Server

The DNS server translates a human-readable domain name into the corresponding IP address of the web server.

Procedure:

1. Open Server0, which is configured as the DNS server.
2. Navigate to **Services → DNS**.
3. Set the DNS service to **On**.
4. Add the following DNS record.

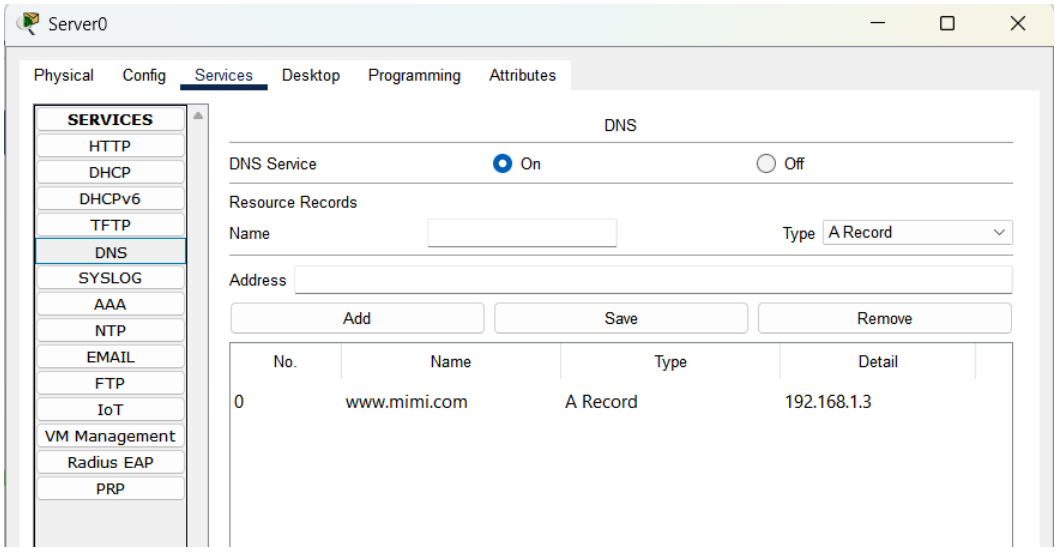
Field	Value
DNS Service	On
Record Type	A Record
Name	www.mimi.com
Address	192.168.1.3

5. Click **Add** to save the record.

This record associates the domain name www.mimi.com with the web server's IP address, 192.168.1.3.

The DNS configuration can be verified in the DNS service window.

Screenshot 2: DNS service showing the configured A record



6.4 Configuring the Web Server

The web server stores and serves the webpage requested by client PCs through HTTP.

Procedure:

1. Open Server1, which is configured as the web server.
2. Navigate to **Services → HTTP**.
3. Set the HTTP service to **On**.
4. Check that the default webpage, usually index.html, is available.
5. If necessary, edit the webpage to display a suitable message, such as:

Welcome to Mimi's Website!

6. Save the changes if the editor provides a save option.

The web server is now configured to respond to HTTP requests from client devices.

Note: HTTPS is not required for this basic HTTP experiment.

6.5 Configuring the Client PCs

Both PCs must be configured to use the DNS server.

For PC0:

- IP address: 192.168.1.10
- Subnet mask: 255.255.255.0
- DNS server: 192.168.1.2

For PC1:

- IP address: 192.168.1.11
- Subnet mask: 255.255.255.0
- DNS server: 192.168.1.2

Once configured, both PCs can send DNS queries to Server0 to resolve the domain name of the web server.

7. Testing and Verification

The network configuration is tested using the ping command and domain name resolution. These tests help verify that the devices can communicate and that the DNS server returns the expected IP address.

7.1 Testing Connectivity to the DNS Server

On a client PC, open **Desktop → Command Prompt** and execute:

ping 192.168.1.2

Successful replies indicate that the client can communicate with the DNS server.

7.2 Testing Connectivity to the Web Server

Execute the following command:

ping 192.168.1.3

Successful replies indicate that the web server is reachable over the network.

The recorded ping tests to both server IP addresses showed four replies received and zero packet loss.

7.3 Testing DNS Resolution

Execute the following command:

ping www.mimi.com

The DNS server should resolve the domain name to 192.168.1.3, after which the PC attempts to ping that address.

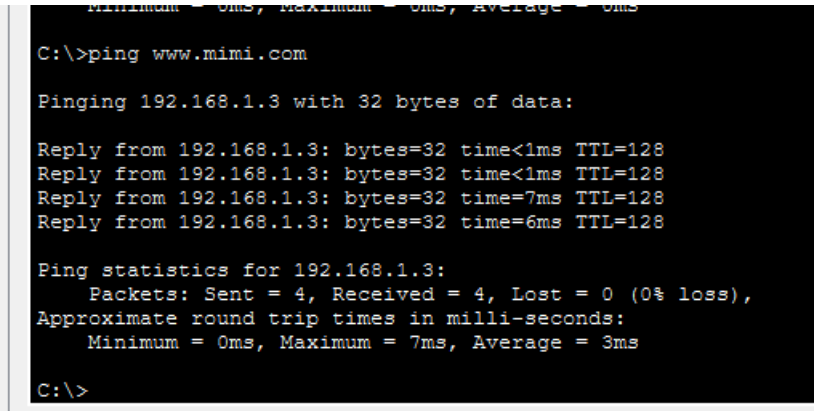
The observed output resolved www.mimi.com to 192.168.1.3 and returned successful replies with zero packet loss.

7.4 Summary of Test Results

Sr. No.	Test	Expected Result	Observed Result
1	Ping DNS server (192.168.1.2)	Successful replies	Successful; 0% packet loss
2	Ping web server (192.168.1.3)	Successful replies	Successful; 0% packet loss
3	Ping domain	Resolves to	Domain resolved correctly; successful

	(www.mimi.com)	192.168.1.3	replies
4	Open http://www.mimi.com in the web browser	Webpage displayed	To be verified in the browser

Screenshot 3: Command Prompt showing successful DNS resolution and ping results

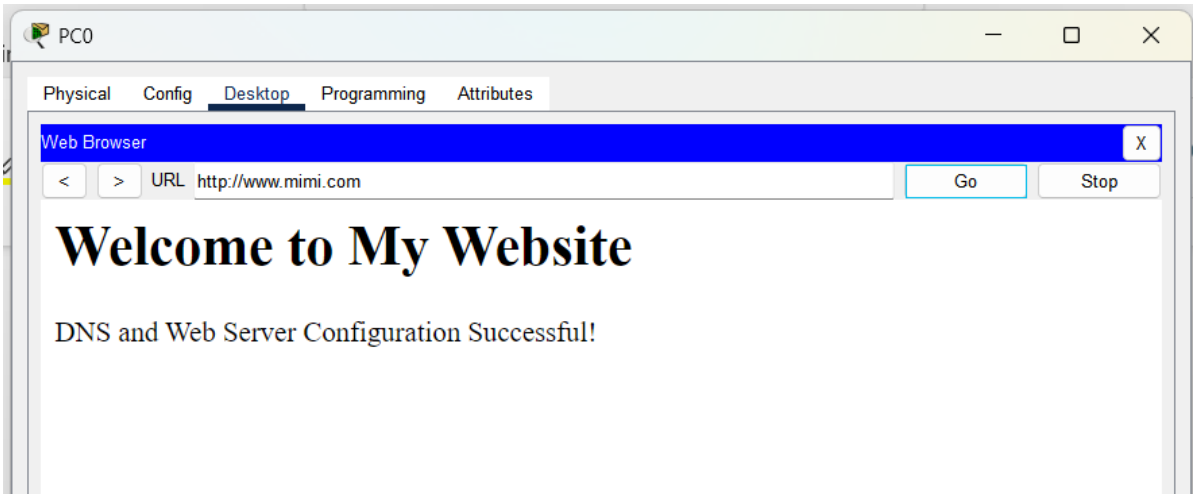


7.5 Accessing the Website Through a Browser

To perform the final application-level test:

1. Open PC0.
2. Navigate to **Desktop → Web Browser**.
3. Enter the URL http://www.mimi.com.
4. Click **Go**.
5. Verify that the webpage hosted on the web server is displayed.

If the webpage loads successfully, it confirms that the DNS resolution and HTTP service work together to provide access through the domain name.



8. Working Principle

DNS and HTTP perform different roles in accessing a website.

When a user enters http://www.mimi.com into a browser, the client first needs to resolve the domain name to an IP address. The PC sends a DNS query to the DNS server at 192.168.1.2. The DNS server checks its records and returns 192.168.1.3, the IP address of the web server.

The browser then sends an HTTP request to the web server at 192.168.1.3. If the HTTP service is enabled and the requested webpage is available, the server returns the webpage to the client.

The overall process is:

Domain name → DNS query → IP address resolution → HTTP request → Webpage response

This demonstrates how DNS makes web access easier by allowing users to use domain names instead of remembering numerical IP addresses.

9. Result

The network topology was created successfully in Cisco Packet Tracer, and the devices were assigned IP addresses according to the planned addressing scheme.

The DNS server was configured with an A record mapping www.mimi.com to 192.168.1.3. Connectivity to both servers was verified using the ping command, and the domain name resolved correctly to the web server's IP address.

The final browser test should be completed to confirm that the webpage can be accessed through http://www.mimi.com.

10. Conclusion

This experiment demonstrates the configuration and operation of DNS and HTTP services in a simulated computer network. Static IP addressing enabled communication between the PCs, DNS server, and web server. The DNS service successfully translated the domain name www.mimi.com into the IP address 192.168.1.3, and the successful ping tests confirmed connectivity to the servers.

The experiment illustrates the importance of DNS in domain name resolution and HTTP in delivering webpages to client devices. Accessing the website

through the browser provides the final verification that both services are configured correctly and functioning together.
